

Comprehensive Analysis: T0 Theory and Matsas et al.
(2024) A Complete Comparative Study of Fundamental
Constants Reduction From Spacetime Structure to
Geometric Unity

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Abstract

This document provides a comparative analysis relating T0 theory, which traces the physical constants back to the one geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$, to the paper by Matsas et al. (2024): "The number of fundamental constants from a spacetime-based perspective" (Scientific Reports, DOI: 10.1038/s41598-024-71907-0). The paper by Matsas et al. proposes a resolution of the long-standing Duff-Okun-Veneziano controversy by demonstrating that in relativistic spacetimes only one fundamental constant (connected to the time unit) is necessary. T0 theory complements this approach through a geometric reduction to the one parameter ξ , from which the physical constants—including dimensionless ones like the fine structure constant α —are derived; SI anchors such as m_e serve only to convert into units, not to derive. The analysis includes mathematical derivations, philosophical reflections and experimental proposals, and shows where both approaches meet on the way toward a unified understanding of quantum mechanics, quantum field theory, and relativity. Many core ideas—particularly the derivability of masses via Compton wavelength and the interpretation of constants like c , G , and k_B as conversion factors—overlap significantly between the two frameworks. The parallels are this document's own classification, not a statement by Matsas et al.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction: The Quest for Fundamental Constants

Historical Context

The question "How many fundamental constants does physics truly need?" has been a central philosophical and practical concern since the early 20th century. When Max Planck introduced his natural units in 1899, he proposed that c , G , and $\hbar$ might represent fundamental scales of nature. However, the debate intensified with the development of quantum field theory and the standardization of measurement systems.

The Duff-Okun-Veneziano (DOV) controversy, initiated in the early 2000s, crystallized different perspectives on this question:

- **Michael Duff:** Argued that only dimensionless constants (like α , mass ratios) are truly fundamental since dimensional constants can be set to 1 by choice of units.
- **Lev Okun:** Maintained that dimensional constants (c , $\hbar$, G) are fundamental because they relate different physical dimensions.
- **Gabriele Veneziano:** Took an intermediate position, suggesting the answer depends on the theoretical framework.

Matsas et al. Resolution

The paper by Matsas et al. (2024) proposes a resolution by showing that the number of fundamental constants is **framework-dependent**:

- In Galilean (non-relativistic) spacetime: **three** constants are needed
- In relativistic spacetime (special relativity): **one** constant suffices
- In general relativistic spacetime: **zero or one**, depending on interpretation

Their key insight: In relativistic spacetimes, a single time unit (operationally defined by real clocks) suffices to express all observables. Space, mass, and other quantities become derivable rather than independent.

T0 Theory's Geometric Reduction

T0 theory pursues the reduction even further by grounding physics in pure geometry. The central claim:

Key Result

The physical constants are traced back to a single geometric parameter:

$$\xi = \frac{4}{3} \times 10^{-4}$$

which represents the ratio between tetrahedral and spherical packing in spacetime at the Planck scale.

This parameter ξ is not fitted to experimental data but emerges from fundamental geometric principles related to the most efficient packing structures in 3D space. From ξ , T0 theory derives (integer structural choices are motivated settings, and SI anchors such as m_e serve to convert into units):

1. All particle masses (electron, muon, proton, etc.)
2. The speed of light c
3. The gravitational constant G
4. The Planck constant $\hbar$ only as a scaling $\hbar \sim \sqrt{\xi} \times E_{\text{scale}} \times T_{\text{scale}}$; a derivation of its numerical value from ξ is not available
5. The fine structure constant α
6. Coupling constants (for the strong and weak couplings no viable relation exists yet) and mass hierarchies

Purpose of This Analysis

Both works pursue the common goal of minimizing the number of “fundamental” physical constants, but from different starting points:

- **Matsas et al.:** Start from spacetime structure and show operationally that in relativistic spacetimes a single unit (time, defined by real clocks) suffices to express all observables.
- **T0 theory:** Goes one step further and reduces everything to a single geometric parameter ξ , whereby even the speed of light c and gravitational constant G are considered derived quantities.

This comprehensive analysis explores:

1. Conceptual overlaps between the approaches

2. Where T0 theory connects to and extends Matsas's framework
3. Mathematical derivations of constants from ξ
4. The flexibility of choosing different starting parameters
5. Philosophical implications for our understanding of fundamental physics
6. Experimental verification proposals
7. The unification of quantum mechanics, quantum field theory, and relativity

.2 Conceptual Overlaps and Convergences

Reduction of the Number of Fundamental Constants

Both frameworks markedly reduce the number of independent inputs:

- **Matsas et al.:** Arrive at exactly **one** constant (time unit) in relativistic spacetimes. By showing that space ($x = c \cdot t$) and mass ($m = \hbar/(c\lambda)$) are derivable from time measurements, they reduce the standard three-constant framework to a single input.
- **T0 theory:** Works with the **one parameter** ξ (plus motivated integer structural choices; SI anchors serve for conversion), whereby units like time or length become secondary consequences of the geometric structure. Even the time unit emerges from ξ through the Planck time $t_P = \sqrt{\hbar G/c^5}$ where both G and c are ξ -derived.

The philosophical common ground: Both reject the notion that multiple independent constants are "fundamental" in any deep sense. They are either:

1. Conversion factors between humanly convenient units (Matsas perspective)
2. Manifestations of a single underlying geometric structure (T0 perspective)

Derivability of $G, c, \hbar, k_B$

Both frameworks do not see these constants as fundamental in the traditional sense:

Speed of Light c

- **Matsas (Eq. 20):** c is a conversion factor between space and time measurements, with value determined by operational protocols (e.g., radar distance measurement).
- **T0 theory:** c emerges geometrically from the ratio l_P/t_P of Planck scales, which themselves derive from ξ :

$$c^2 \sim \frac{1}{\xi \cdot D_f}$$

where $D_f = 3 - \xi$ is the fractal dimension of spacetime near the Planck scale.

Gravitational Constant G

- **Matsas:** G is derivable indirectly via masses through gravitational laws and Compton wavelength relations. In geometrized units, it can be set to 1.
- **T0 theory:** G is explicitly derived from ξ (detailed in Section .5):

$$G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$$

This connects Planck length $l_P = \sqrt{\hbar G/c^3}$ directly with ξ , showing that gravitational strength is a geometric consequence.

Planck Constant $\hbar$

- **Matsas:** $\hbar$ appears in the Compton wavelength relation $\bar{\lambda} = \hbar/(mc)$ and is part of the reduction process, ultimately set by fixing the time scale.
- **T0 theory:** $\hbar$ scales with $\sqrt{\xi}$, reflecting hierarchical energy-time relationships:

$$\hbar \sim \sqrt{\xi} \times (\text{energy scale}) \times (\text{time scale})$$

This connects quantum action to geometric structure.

Boltzmann Constant k_B

Both works eliminate k_B as fundamental:

- **Matsas:** In the reduction $\text{SI} \rightarrow \text{MKS} \rightarrow \text{MS}$, temperature is absorbed into energy via $E = k_B T$, making k_B a historical convention.
- **T0 theory:** Explicitly treats k_B as a “historical conversion factor” for temperature-energy equivalence, with no fundamental role in the geometric structure.

Spacetime as Starting Point

- **Matsas et al.:** Start with the operational construction of spacetime using clocks and rulers. Time is defined by cesium atomic transitions, space by light propagation. The Unruh-DeWitt detector protocol provides a relativistically covariant definition of time intervals.
- **T0 theory:** Interprets spacetime as “pure ξ -geometry” with a sub-Planck scale:

$$L_0 = \xi \cdot l_P$$

This marks the transition region where quantum effects modify classical spacetime geometry, leading to “spacetime granulation” or fractal structure.

The convergence: Both see spacetime structure (not matter or forces) as the fundamental layer from which everything else emerges.

SI Reform 2019

Both approaches can be related to the 2019 SI reform:

- **Matsas:** The reform (fixing c , $\hbar$, e , k_B) represents a shift from material artifacts to fundamental constants, which fits their operational approach.
- **T0 theory:** Reads the reform as a fixing of conversion factors; the fixed values of constants in the new SI are compatible with the ξ -derived relationships.

.3 Specific Points of Contact between T0 Theory and Matsas et al.

Compton Wavelength as Central Concept

Key Result

The reduced Compton wavelength

$$\bar{\lambda} = \frac{\hbar}{mc}$$

serves as the bridge between mass and spacetime scales in both frameworks.

In Matsas et al. (sections "Two units..." and "Time..."), the reduced Compton wavelength is used to express mass in length or time units:

$$m_e^{\text{MS}} = \frac{\hbar^{\text{MS}}}{c\bar{\lambda}_e}$$

T0 theory takes up this insight: Electron mass m_e is not viewed as an independent parameter but is directly derived from ξ through the quantization formula:

$$m_e = \frac{f(1, 0, 1/2)^2}{\xi^2} \cdot S_{T0}$$

Here:

- $f(1, 0, 1/2)$ is the structure factor of the electron's quantum numbers, encoding its quantum state (not a frequency and not the base winding number $f = 1/\xi$)
- $S_{T0} = 1 \text{ MeV}/c^2$ is a fundamental energy scale in T0 theory
- The $1/\xi^2$ factor reflects the geometric amplification from sub-Planck to observable scales

This fits Matsas's idea that mass is not a fundamental dimension once a base unit (e.g., time) is fixed. T0 offers a geometric mechanism for it.

Reduction to MS or S System

Matsas et al. perform stepwise reduction:

$$\text{SI} \rightarrow \text{MKS} \rightarrow \text{MS} \rightarrow \text{S}$$

where S means "seconds only."

T0 theory complements this by deriving Planck length directly from ξ and m_e :

$$l_P = \sqrt{G} = \frac{\xi}{2\sqrt{m_e}} \times (\text{conversion factors})$$

This provides a *physical justification* for why such reduction is possible: The Planck scale itself emerges from the same geometric structure (ξ) that determines masses. Thus, Planck units are not arbitrary theoretical constructs but reflect the underlying ξ -geometry.

Subsequent conversion to SI units becomes purely conventional and is compatible with Matsas's operational reduction.

Boltzmann Constant as Conversion

Both works eliminate k_B as fundamental:

- **Matsas**: In reduction to MKS, temperature is absorbed via $k_B T = E$
- **T0 theory**: Explicitly identifies k_B as "historical convention" for temperature-energy conversion, with no geometric significance

On this point both agree: Neither framework assigns temperature an independent dimensional status.

.4 The Flexibility of the Base Unit

A particularly interesting common point is the recognition that—once the derivability of all observables is clarified—the choice of the "starting unit" becomes largely conventional.

Matsas Perspective

In relativistic spacetimes, **time** (defined by real clocks) is the natural choice since:

1. Space is derivable: $x = c \cdot t$ (radar measurement protocol)
2. Mass is derivable: $m = \hbar/(c\bar{\lambda})$ via Compton wavelength
3. The Unruh-DeWitt protocol (Eq. 18) provides a covariant time definition

T0 Perspective

Since everything follows from ξ , one could in principle start from:

1. **Length** (l_P): Planck length as fundamental $\rightarrow$ time via $t_P = l_P/c$
2. **Time** (t_P): Planck time as fundamental $\rightarrow$ length via $l_P = c \cdot t_P$
3. **Energy** (S_{T0}): Energy scale as fundamental $\rightarrow$ masses via quantization
4. **Directly from ξ** : Geometric parameter $\rightarrow$ all scales simultaneously
5. **Fine structure constant α** : Dimensionless constant $\rightarrow$ everything else via closed formula chain (see Section .6)

Convergence on Flexibility

This corresponds to an extension of Duff's flexible attitude (no fixed number of standards), but with a crucial difference:

Insight .4.1. The geometric anchor

While Matsas shows operational flexibility in choosing units, T0 provides a *geometric anchor* ξ that ensures all choices lead to the same physics. The flexibility is not arbitrary but constrained by the closed mathematical structure.

.5 Mathematical Derivations

Derivation of the Fine Structure Constant

The fine structure constant $\alpha \approx 1/137.036$ is perhaps the most precisely measured dimensionless constant in physics. T0 theory traces it back to a geometric origin.

Matsas Treatment

Matsas et al. recognize α as a physically significant dimensionless parameter (in Duff's sense) that need not be reduced to a fundamental constant since it is comparable without additional standards. They use it in electrodynamics for reducing the ampere unit (section "Recovering the MKS system"):

$$\alpha = \frac{e^2 k_e}{\hbar c}$$

This helps convert electrical quantities into mechanical units.

However, Matsas does not derive α from more fundamental principles—it remains an input parameter.

T0 Derivation

T0 theory explicitly derives α from the geometric parameter ξ :

Key Result

$$\alpha = \xi \cdot E_0^2$$

where $E_0 = \sqrt{m_e \cdot m_\mu / K_{\text{frak}}} = 7.398 \text{ MeV}$ is a fundamental energy scale (geometric mean of electron and muon masses, 7.348 MeV, fractally corrected with $K_{\text{frak}} = 1 - 100\xi$).

Step-by-step derivation:

1. Start with $\xi = \frac{4}{3} \times 10^{-4}$ from tetrahedral geometry
2. Derive electron mass: $m_e = 0.511 \text{ MeV}/c^2$ from quantization formula
3. Derive muon mass: $m_\mu = 105.66 \text{ MeV}/c^2$ similarly
4. Compute the geometric mean energy and its fractal correction:

$$E_0 = \sqrt{\frac{m_e \cdot m_\mu}{K_{\text{frak}}}} = \frac{\sqrt{0.511 \times 105.66}}{\sqrt{K_{\text{frak}}}} \text{ MeV} = \frac{7.348}{\sqrt{K_{\text{frak}}}} \text{ MeV} \approx 7.398 \text{ MeV}$$

5. Apply the relation, with E_0 as a pure number in MeV:

$$\begin{aligned} \alpha &= \xi \cdot E_0^2 = \frac{4}{3} \times 10^{-4} \times (7.398)^2 \\ &= \frac{4}{3} \times 10^{-4} \times 54.73 \approx \frac{1}{137.04} \end{aligned}$$

Physical interpretation:

The fine structure constant represents the strength of electromagnetic interaction. In T0 theory, this strength is determined by:

- The geometric structure parameter ξ (spatial packing efficiency)
- The mass hierarchy $E_0^2 = m_e \cdot m_\mu / K_{\text{frak}}$ (lepton sector structure)

Thus, electromagnetism's strength is not arbitrary but follows from spacetime geometry and mass quantization.

Derivation of the Gravitational Constant

Matsas Treatment

Matsas et al. treat G as a conversion factor transforming masses into space and time units (section “Two units... in Galilean spacetime”). They do not directly derive G but show that in geometrized units (S-system) it can be set to 1 since observables are expressible in time units alone.

Indirectly, G results from the derivability of masses via Compton wavelength and gravitational laws.

T0 Derivation

In T0 theory, G is explicitly derived from ξ :

Key Result

$$G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$$

Derivation:

1. Start with Planck length definition:

$$l_P = \sqrt{\frac{\hbar G}{c^3}}$$

2. In T0 theory, Planck length emerges from sub-Planck scale:

$$l_P = \frac{L_0}{\xi} = \frac{\xi \cdot l_P}{\xi} \implies L_0 = \xi \cdot l_P$$

3. Relating the Planck length to the electron mass via $\bar{\lambda}_e/\sqrt{\alpha}$ does not work:

$$\frac{\bar{\lambda}_e}{\sqrt{\alpha}} = \frac{\hbar}{m_e c \sqrt{\alpha}} \approx 4.5 \times 10^{-12} \text{ m} \neq l_P$$

A derivation of G via $l_P = \bar{\lambda}_e/\sqrt{\alpha}$ therefore does not hold, because the expression gives $4.5 \times 10^{-12} \text{ m}$ instead of $1.6 \times 10^{-35} \text{ m}$.

4. The value of G follows from the key result $G = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$ with C_{conv} as the SI conversion (see Doc. 127):

$$G \approx 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

Physical interpretation:

The gravitational constant's weakness ($G \ll 1$ in Planck units) arises from:

- The ξ^2 factor (very small since $\xi \sim 10^{-4}$)
- The inverse electron mass relationship
- The fractal correction factor K_{frak} from spacetime granulation

Within the T0 framework this yields an interpretation of the hierarchy problem: Gravity is weak because it couples to the sub-Planck geometric structure encoded in ξ^2 .

Derivation of Speed of Light

Key Result

$$c^2 \sim \frac{1}{\xi \cdot D_f}$$

where $D_f = 3 - \xi$ is the effective fractal dimension of spacetime.

Derivation:

1. Spacetime at the Planck scale has fractal structure with dimension:

$$D_f = 3 - \xi \approx 2.99987$$

2. Light propagation in fractal spacetime experiences effective impedance:

$$Z_{\text{eff}} = \xi \cdot D_f$$

3. Speed of light relates to impedance:

$$c^2 = \frac{1}{\epsilon_0 \mu_0} \sim \frac{1}{Z_{\text{eff}}} = \frac{1}{\xi \cdot D_f}$$

4. Numerical evaluation:

$$c^2 = \frac{1}{\frac{4}{3} \times 10^{-4} \times 2.99987} \approx 2.5 \times 10^3$$

so $c \approx 50$ in these units; no bridge to 3×10^8 m/s results from this. The numerical value 299 792 458 m/s is the SI anchor.

Physical interpretation:

In the T0 framework the speed of light is not a fundamental constant; it reflects:

- The geometric packing efficiency ξ (how densely spacetime is structured)
- The fractal dimension D_f (how spacetime deviates from perfect 3D Euclidean space)

Light travels at c because that is the maximum speed at which causal signals can propagate through ξ -structured spacetime.

Scaling of Planck Constant

Key Result

$$\hbar \sim \sqrt{\xi} \times E_{\text{scale}} \times T_{\text{scale}}$$

Derivation:

1. Planck constant relates energy and frequency: $E = \hbar \omega$
2. In T0 theory, energy scales with ξ hierarchy:

$$E_{\text{scale}} = S_{T0} = 1 \text{ MeV}/c^2$$

3. Time scales inversely with energy:

$$T_{\text{scale}} = \frac{\hbar}{E_{\text{scale}}}$$

4. Self-consistent relation:

$$\hbar = E_{\text{scale}} \cdot T_{\text{scale}} \sim \sqrt{\xi} \times \frac{l_{pc}^2}{c} = \sqrt{\xi} \times l_{pc}$$

5. This expression is dimensionally incomplete: $[l_{pc}] = \text{m}^2/\text{s}$, $[\hbar] = \text{kg m}^2/\text{s}$; a mass is missing. Numerically $\sqrt{\xi} l_{pc} = 0.01155 \cdot 1.616 \times 10^{-35} \text{ m} \cdot 2.998 \times 10^8 \text{ m/s} = 5.6 \times 10^{-29} \text{ m}^2/\text{s}$; no stated factor leads to $\hbar = 1.055 \times 10^{-34} \text{ J s}$.

Physical interpretation:

The Planck constant represents the quantum of action. The scaling $\hbar \sim \sqrt{\xi} \times E_{\text{scale}} \times T_{\text{scale}}$ links it to the $\sqrt{\xi}$ factor reflecting the energy-time hierarchy. The form $\sqrt{\xi} l_{pc}$ built from Planck length and speed of light, however, lacks a mass and does not yield the numerical value of $\hbar$; this chain does not provide a derivation of $\hbar$ from ξ .

.6 Alternative Formulations of T0 Theory: Closed Derivation Chain

The Crucial Condition

Key Result

A closed chain of formulas

One cannot arbitrarily switch in T0 theory between ξ , α , a mass scale, or a measured constant as the "fundamental" parameter *without* knowing a completely closed, consistent chain of derivation formulas. Only when this chain is mathematically exact and internally consistent does the physics remain identical, regardless of which point one starts from.

The Standard Chain

In current T0 theory, the chain is constructed as:

$$\xi \rightarrow m_e \quad (\text{via mass quantization formula})$$

$$m_e = \frac{f(1, 0, 1/2)^2}{\xi^2} \cdot S_{T0} \quad (1)$$

$$m_e \rightarrow E_0 = \sqrt{m_e \cdot m_{\mu} / K_{\text{frak}}} \quad (\text{geometric mean energy, fractally corrected}) \quad (2)$$

$$\xi, E_0 \rightarrow \alpha = \xi \cdot E_0^2 \quad (3)$$

$$m_e, \xi \rightarrow G = \frac{\xi^2}{4m_e} \times \text{conversion factors} \quad (4)$$

$$G \rightarrow l_p = \sqrt{G} \quad (5)$$

This chain is closed and therefore allows multiple equivalent starting points.

Alternative 1: Fine Structure Constant as Fundamental

This is particularly attractive since α is very precisely measured ($\alpha^{-1} = 137.035999177(21)$).

Reversed chain starting from α :

1. Start with measured $\alpha \approx 1/137.036$

2. Derive ξ from inverted relation:

$$\xi = \frac{\alpha}{E_0^2}$$

But this requires knowing $E_0 = \sqrt{m_e \cdot m_\mu / K_{\text{frac}} \dots}$

3. Solve self-consistently:

- Assume ξ ansatz from geometry ($\xi \sim 1.33 \times 10^{-4}$)
- Compute m_e, m_μ from quantization formulas
- Verify $\alpha = \xi \cdot E_0^2$ matches experiment

4. All other constants follow:

$$\alpha \rightarrow m_e \text{ (via inverse quantization)} \quad (6)$$

$$m_e, \xi \rightarrow G \rightarrow l_p \quad (7)$$

$$\xi, D_f \rightarrow c \quad (8)$$

This works only because E_0 and S_{T0} are fixed by the same geometric logic. The closure of the chain ensures consistency.

Alternative 2: Measured Constant as Starting Point

Theoretically possible but less favourable since G and m_e are less precisely known than α .

Example: Starting from electron mass

1. Measured: $m_e = 0.51099895000(15) \text{ MeV}/c^2$
2. Invert quantization formula:

$$\xi = \frac{f(1, 0, 1/2)}{\sqrt{m_e / S_{T0}}}$$

3. Compute α :

$$\alpha = \xi \cdot \frac{m_e \cdot m_\mu}{K_{\text{frac}} (1 \text{ MeV})^2}$$

i.e. $\xi \cdot 53.99 / 0.98667 = 1 / 137.06$.

4. Verify against experiment
5. Derive G and c as before ($\hbar$ only as a scaling, see above)
Again, the chain must converge on the same value of ξ (or α) to maintain consistency.

.7 The Unification of QM, QFT, and RT

The Central Unification Principle

Key Result

Fundamental Unification

The consistent reduction to *only one* fundamental input (time unit for Matsas or geometric parameter ξ/α for T0) suggests a unification of the three great pillars of modern physics:

- Quantum Mechanics (QM)
- Quantum Field Theory (QFT)

- Relativity Theory (RT)

Integration of Quantum Mechanics

Standard QM: Postulates $\hbar$ as fundamental, with masses as input parameters.

Matsas approach: Masses become derivable via Compton wavelength $\bar{\lambda} = \hbar/(mc)$, with $\hbar$ fixed by time standard.

T0 approach: Masses emerge from ξ ; for $\hbar$ only a scaling is available:

- Masses: $m_i = f_i^2 / \xi^2 \cdot S_{T0}$ (quantization formula)
- $\hbar$: scaling $\hbar \sim \sqrt{\xi} \times E_{\text{scale}} \times T_{\text{scale}}$ (action quantum; the form $\sqrt{\xi} l_{\text{pc}}$ lacks a mass)

Result: QM becomes a geometric theory where quantum properties (discrete masses, quantized action) reflect underlying spacetime structure.

Integration of Quantum Field Theory

Standard QFT: Coupling constants (α , strong coupling α_s , weak coupling g_w) are free parameters fitted to experiment.

Matsas approach: Does not address coupling constants explicitly, but the dimensional reduction removes some arbitrariness.

T0 approach: Coupling constants become geometric:

- $\alpha = \xi \cdot E_0^2$ (electromagnetic coupling)
- Strong and weak coupling: scalings of the form $\xi \cdot (m/m_e)^2$ do not yield them. With $m_{\text{QCD}} = 200\text{--}330$ MeV they give values of 20–56 instead of $\alpha_s \approx 0.118$, and with $m_W = 80.4$ GeV the value 3.3×10^6 for g_w^2 .

Result **[S]**: QFT couplings would no longer be arbitrary but determined by mass hierarchies and ξ ; for the strong and weak couplings no viable relation exists yet. For the “landscape problem” of string theory (many possible coupling values) this would give an answer: Only one set of couplings would be geometrically consistent.

Integration of Relativity Theory

Standard RT: Postulates spacetime metric with c as fundamental speed and G as gravitational coupling.

Matsas approach: Spacetime structure (operationally defined by clocks and rulers) becomes the foundation. c is a conversion factor, G is derivable.

T0 approach: Spacetime metric itself emerges from ξ -geometry:

- Metric signature: $(+, -, -, -)$ from tetrahedral packing symmetry
- c : Maximum causal speed from $c^2 \sim 1/(\xi \cdot D_f)$
- G : Gravitational coupling from $G \sim \xi^2 / m_e$
- Planck scale: $l_p = \xi / (2\sqrt{m_e})$ (quantum gravity threshold)

Result: RT becomes a low-energy effective theory of ξ -structured spacetime. The Einstein equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

represent the geometric response of ξ -spacetime to matter-energy distribution.

The Unified Picture

In standard physics, QM, QFT, and RT appear as separate theories with different fundamental constants:

- QM: $\hbar$, masses
- QFT: α , coupling constants
- RT: c , G

The reduction to a single starting variable (time for Matsas, ξ for T0) reveals that this separation is artificial:

Insight .71. All three areas are manifestations of the same underlying geometric structure.

Physical interpretation in T0:

- The sub-Planck scale $L_0 = \xi \cdot l_P$ marks the transition region where quantum effects modify spacetime geometry ("spacetime granulation").
- The geometric fixation of α is the starting point for tracing electromagnetic, weak, and strong interactions as well as gravitation back to the same ξ parameter **[S]** (for the weak and strong couplings no viable relation exists here yet).
- The derivation of G from the same geometry that also determines quantum masses connects quantum and gravitational theory through the same parameter.

Matsas et al. lay the foundation with their operational reduction to one time unit: Once space, mass, and charge are derivable from time, the apparent differences between relativistic mechanics, quantum mechanics, and electrodynamics disappear.

T0 theory carries this thought further by tracing even this one time unit (or its scale) back to a purely geometric number (ξ or α).

In the T0 framework, the unification of quantum theory and gravitation is pursued not by adding new fields or dimensions but through *reduction* to a single fundamental input.

.8 Philosophical Reflections on Fundamental Constants

What Makes a Constant "Fundamental"?

The DOV controversy and Matsas et al. resolution raise basic philosophical questions:

1. **Operational Definition:** Is a constant fundamental if it is directly measurable with operational protocols? (Matsas perspective)
2. **Dimensional Independence:** Is a constant fundamental if it connects independent physical dimensions? (Okun perspective)
3. **Mathematical Necessity:** Is a constant fundamental if it cannot be eliminated from physical equations? (Duff perspective)
4. **Geometric Origin:** Is a constant fundamental if it represents pure geometric structure without dimensional content? (T0 perspective)

T0 Resolution: Hierarchy of Fundamentalness

T0 theory suggests a hierarchy:

1. **Most fundamental:** ξ (pure geometric ratio, dimensionless)
2. **Derived fundamental:** α (dimensionless constant from ξ)

3. **Geometric scales:** l_P, t_P, m_P (Planck units from ξ and m_e)
4. **Conventional units:** $c, \hbar, G, k_B$ (conversion factors between human-convenient units)

The Role of Geometry vs. Convention

Matsas insight: Many "constants" are merely conventional conversion factors reflecting our choice of measurement standards (meter, second, kilogram).

T0 extension: Even the "natural" scales (Planck length, Planck time) are not arbitrary but reflect the geometric structure of spacetime encoded in ξ .

Convergence: Both reject the notion that physics contains multiple independent "fundamentals." Both start from one fundamental structure (spacetime for Matsas, ξ -geometry for T0) from which the rest follows.

Implications for Fundamental Physics

Insight .8.1. Consequence for the landscape problem [S]

If the constants do derive from one geometric parameter, there is no "landscape" of possible universes with different constant values. Our universe would then have its specific constants because they are the only solution of the ξ -geometric structure.

Insight .8.2. An interpretation of fine-tuning [S]

Constants would then not be "fine-tuned" for life by coincidence or anthropic selection, but would express a self-consistent geometric structure. The universe could not be "otherwise" without violating geometric consistency.

Insight .8.3. Unification without extra dimensions [S]

A unified description need not bring forces together by adding extra dimensions or supersymmetry. It can also be pursued by showing that physics derives from one geometric principle (ξ -structure). QM, QFT, RT then appear as different aspects of the same geometry.

Historical References on Physical Constants

- **Planck, M.** (1899). Über irreversible Strahlungsvorgänge. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, 440–480.
Introduction of Planck constants and natural units. In T0 framework, Planck scales derive from ξ rather than being fundamental.
- **Duff, M. J.** (2004). Comment on time-variation of fundamental constants. *Physical Review D*, 70, 087505. DOI: <https://doi.org/10.1103/PhysRevD.70.087505>.
Duff's perspective on dimensionless constants as truly fundamental. Fits T0 insofar as α can serve as an alternative starting point to ξ .
- **Okun, L. B.** (1989). The concept of mass. *Physics Today*, 42(6), 31–36. DOI: <https://doi.org/10.1063/1.881171>.
Discussion of mass as fundamental constant. Contrasted with T0 derivation of all masses from ξ via quantization formula.

T0 Theory Documents

- 008_T0_xi-und-e_En.pdf: Connection between ξ and elementary charge e . Shows geometric origin of electromagnetic coupling, adding an explicit charge derivation to the comparison with Matsas.

- 009_T0_xi_origin_En.pdf: Origin of ξ from tetrahedral packing geometry. Introduces fractal dimension $D_f = 3 - \xi$ explaining spacetime structure near Planck scale.
- 042_xi_parameter_particles_En.pdf: ξ -based particle mass spectrum. Shows how electron, muon, tau, quarks derive from same quantization formula. Complements Matsas reduction with quantum field theory implications and experimental verification.
- 011_T0_Feinstruktur_En.pdf: Detailed derivation of fine structure constant $\alpha = \xi \cdot E_0^2$ from geometric principles.
- 012_T0_Gravitationskonstante_En.pdf: Derivation of gravitational constant G from ξ , explaining gravity-quantum hierarchy.

Related Experimental and Theoretical Work

- **Koide, Y.** (1982). Quark and lepton masses speculated from a subquark model. *Lettere al Nuovo Cimento*, 34(7), 201–205.
Koide formula for charged lepton masses. An independent empirical mass relation that forms a parallel to the T0 quantization structure.
- **CODATA Recommended Values** (2018). *Reviews of Modern Physics*, 93, 025010. DOI: <https://doi.org/10.1103/RevModPhys.93.025010>.
Precision values of fundamental constants for comparison with T0 predictions.

This analysis sets out the points of contact between operational reduction (Matsas et al.) and geometric reduction (T0 theory), together with the T0 thesis that physics at its deepest level is geometry.